Ernesto Camilo Zuleta Suarez, PhD
Curriculum Vitae
eczuletas@gmail.com | Google scholar

PROFILE

Energy scientist with expertise in electrochemical energy storage, redox flow batteries, electrochemical diagnostics, and molecular synthesis of redox-active materials. More than five years of postdoctoral research experience at Oak Ridge National Laboratory and the University of Tennessee developing advanced sodium-based energy storage systems for grid-scale applications.

EDUCATION

Jan. 2014 – Dec. 2019 | **Ph.D. in Energy Science and Engineering**, The University of Tennessee, Bredesen Center for Interdisciplinary Research and Graduate Education, Knoxville, TN, US.

Jan. 2010 – Jun. 2013 | **M.Sc. in Engineering (Chemical Engineering)**, Universidad de Antioquia, Medellín, Colombia.

Jan. 2000 – Jun. 2007 | **B.Sc. in Agro-industrial Engineering**, Universidad Popular del Cesar, Valledupar, Colombia.

PUBLICATIONS

Peer-reviewed journal articles

Ernesto C. Zuleta S., V. Sethuraman, M. L. Lehmann, G. Goenaga, G. Yang, L. Cheng, E. C. Self. 2026. Solvent-based synthesis and characterization of sodium thiophosphate catholytes for nonaqueous redox flow batteries. *Chemical Engineering Journal*, 173687.

M. Escamilla, **Ernesto C. Zuleta**, H. K. Davis, J. Johnson, E. Pentzer, T. Zawodzinski. 2024. Synthesis of Alkoxy-TEMPO Aminoxyl Radicals and Electrochemical Characterization in Acetonitrile for Energy Storage Applications. *Journal of The Electrochemical Society*, 171, 040533.

Ernesto C. Zuleta, J. J. Bozell. 2021. Alkylation of monomeric, dimeric, and polymeric lignin models through carbon–hydrogen activation using Ru-catalyzed Murai reaction. *Tetrahedron*, 100, 132475.

Ernesto C. Zuleta, G. A. Goenaga, T. A. Zawodzinski, T. Elder, J. J. Bozell. 2020. Deactivation of Co-Schiff Base Catalysts in the Oxidation of para-Substituted Lignin Models for the Production of Benzoquinones. *Catalysis Science & Technology*, 10, 403–413.

L. M. Baena, **Ernesto C. Zuleta**, J. A. Calderón. 2018. Evaluation of the stability of polymeric materials exposed to palm biodiesel and biodiesel–organic acid blends. *Polymers*, 10, 511.

Ernesto C. Zuleta, L. A. Rios, P. N. Benjumea. 2012. Oxidative stability and cold flow behavior of palm, sacha-inchi, jatropha, and castor oil biodiesel blends. *Fuel Processing Technology*, 102, 96–101.

Ernesto C. Zuleta, L. Baena, L. Rios, J. A. Calderón. 2012. The oxidative stability of biodiesel and its impact on the deterioration of metallic and polymeric materials: a review. *Journal of the Brazilian Chemical Society*, 23, 2159–2175.

Ernesto C. Zuleta, J. A. Calderón Gutiérrez, L. A. Rios. 2012. Study of the oxidative stability of biodiesel from palm oil in contact with metallic and polymeric automotive materials. *Ingeniería y Competitividad*, 2, 83–90.

Ernesto C. Zuleta, A. Franco, L. A. Rios. 2009. Production of palm oil biodiesel with heterogeneous basic catalysts compared to conventional homogeneous catalysts. *Energética*, 42, 45–52.

Ernesto C. Zuleta, M. J. Bastidas Barranco, L. C. Díaz Muegue, J. Bonet Ovalle. 2007. Preparation of biodiesel by transesterification of crude palm oil (Elaeis guineensis) with ethanol. *Energética*, 38, 47.

I. R. Avendaño Gómez, L. C. Díaz Muegue, E. Meza Daza, **Ernesto Camilo Zuleta Suarez**, M. Mantilla Morales. 2007. Study of the obtention of epoxidized palm olein by peroxyacetic acid formed in situ. *Revista Científica Documentos de Ingeniería*, 6, 69–75.

Book chapters

J. J. Bozell, S. Chmely, W. Hartwig, R. Key, N. Labbé, P. Venugopal, **Ernesto Zuleta**. 2018. Lignin isolation methodology for biorefining, pretreatment and analysis. In: *Lignin Valorization: Emerging Approaches*, Ed. G. T. Beckham, The Royal Society of Chemistry, UK, pp. 21–61.

L. Ríos, S. Cardona, W. Pérez, **Ernesto Zuleta**. 2012. Oxidative stability of biodiesel. In: *Alternative Production of Biodiesel: Improvement of Properties and Assessment of Glycerin*, Universidad de Antioquia, Medellín, Colombia, pp. 49–64.

INVITED PRESENTATIONS

Ernesto C. Zuleta Suarez. "Hydrogen: An Energy Alternative with Many Colors and Shades" ("Hidrógeno: Una alternativa energética con muchos colores y matices"). Invited webinar speaker, Universidad Autónoma de Bucaramanga (UNAB), Bucaramanga, Colombia, Oct. 2023. (virtual)

ORGANIZED SESSIONS AND SYMPOSIA

Organizer, First CRC–CABLE Bioeconomy Seminar, "Bioeconomy in East Tennessee." Center for Renewable Carbon, The University of Tennessee, Knoxville, 2018.

PRESENTATIONS

Ernesto C. Zuleta Suarez, G. Goenaga, E. Self. Evaluation of Anode Designs for Sodium Metal Hybrid Redox Flow Batteries. 249th ECS Meeting, 2026, Seattle, WA. (oral)

Ernesto C. Zuleta Suarez, M. Lehmann, G. Yang, E. Self. Sodium Polysulfides as Catholytes for Redox Flow Batteries. 245th ECS Meeting, 2024, San Francisco, CA. (oral)

Ernesto C. Zuleta Suarez et al. Synthesis and electrochemical characterization of alkoxy-TEMPO radicals for energy storage. ACS Fall Meeting, 2022, Chicago, IL. (oral)

Ernesto C. Zuleta Suarez, L. A. Rios, L. Baena, J. A. Calderón. Study of the oxidative stability of Colombian biodiesel exposed to metallic and polymeric materials in long-term tests. V Congreso Internacional de Ciencia y Tecnología de los Biocombustibles (CIBSCOL), 2012, Bucaramanga, Colombia.

L. Baena, **Ernesto Zuleta**, J. A. Calderón. Evaluation of impact on polymer properties of palm oil biodiesel mixed with fatty acids. CIBSCOL, 2012, Bucaramanga, Colombia.

Ernesto Zuleta, L. A. Rios. Oxidative stability of biodiesel from palm oil in contact with metallic and polymeric materials. 14th AOCS Latin American Congress and Exhibition on Fats and Oils, 2011, Cartagena, Colombia. (poster)

L. A. Rios, **Ernesto Camilo Zuleta Suarez**, J. A. Calderón. Oxidative stability of biodiesel from palm oil in contact with metallic and polymeric materials. Euro Fed Lipid Congress, 2011.

Ernesto Camilo Zuleta Suarez, L. A. Rios, G. Bonilla. Oxidative stability and cold flow behavior of palm, sacha-inchi, jatropha, and castor oil biodiesel blends. XXIX Congreso Latinoamericano de Química (CLAQ), 2010, Colombia.

RESEARCH AND WORK EXPERIENCE

Jul. 2023 – Jul. 2026 | **Oak Ridge National Laboratory**, Energy Storage and Conversion Group, Oak Ridge, TN
Postdoctoral Research Associate

- Developed and evaluated sodium hybrid redox flow battery architectures and interfaces through advanced electrochemical diagnostic techniques, including GEIS/PEIS and equivalent-circuit modeling of Na|β''-alumina solid-electrolyte cells, to identify performance limitations, optimize interfacial design, and improve long-term electrochemical stability.
- Synthesized sodium thiophosphate and polysulfide compounds and characterized them by ^{31}P-NMR, UV-Vis, and Raman spectroscopy.
- Performed electrochemical characterization of catholyte materials using cyclic voltammetry, coin-cell, and flow-cell testing for nonaqueous redox flow batteries.

Nov. 2020 – Jun. 2023 | **The University of Tennessee**, Dept. of Chemical and Biomolecular Engineering, Knoxville, TN
Postdoctoral Researcher, Zawodzinski Group

- Evaluated electrochemical performance of organic redox-active molecules in aqueous systems and tested membrane materials in full-cell configurations.
- Applied EPR spectroscopy to study molecular dynamics in deep eutectic solvents.

Jan. 2014 – Aug. 2019 | **The University of Tennessee**, Center for Renewable Carbon, Knoxville, TN
Graduate Research Assistant

- Synthesized and characterized lignin-derived model compounds via catalytic methods; studied reaction mechanisms and catalyst deactivation pathways.
- Performed analytical characterization using HPLC and NMR spectroscopic techniques.

Aug. 2010 – Jun. 2013 | **Universidad de Antioquia**, Medellín, Colombia
Research Assistant

- Developed strategies to mitigate biodiesel corrosivity and its impact on automotive components.
- Improved the oxidative stability and low-temperature flow properties of biodiesel from natural oils.

Jul. 2005 – Oct. 2007 | **Universidad Popular del Cesar, Valledupar, Colombia**
Research Assistant

- Studied the production of epoxidized palm olein via peroxyacetic acid formed in situ.
- Analyzed preparation of activated carbons from carbonaceous precursors of the Cesar bituminous coal basin and agricultural biomass.

TEACHING EXPERIENCE

Jan. 2010 – Dec. 2012 | **Universidad de Antioquia, Faculty of Engineering, Medellín, Colombia**
Lecturer and Teaching Assistant

- Teaching Assistant, Heat Transfer Laboratory (Dept. of Chemical Engineering): conduction, convection, and heat-exchanger experiments; instrumentation, data acquisition, and laboratory report writing.
- Lecturer, Statistics and Design of Experiments (undergraduate Chemical Engineering): hypothesis testing, regression, factorial design, and analysis of experimental data.
- Instructor and statistical consultant, Design and Formulation of Foods (Faculty of Pharmaceutical Chemistry): applied experimental design and formulation optimization.

MENTORSHIP

2023 | **Research mentor**, The University of Tennessee, Knoxville, TN
Hannah K. Davis, undergraduate student.
Jacob Johnson, undergraduate student.

SERVICE AND OUTREACH

Aug. 2023 – Aug. 2024 | Board member of Oak Ridge National Laboratory Postdoc Association (ORPA)

LEADERSHIP ROLES

Aug. 2023 – Aug. 2024 | Social and Well-being Chair, Oak Ridge National Laboratory Postdoc Association

Aug. 2017 – Apr. 2018 | Student Delegate, Consortium for Advanced Bioeconomy Leadership Education (CABLE), The Ohio State University

PROFESSIONAL DEVELOPMENT

Digital Twins and Advanced Simulation in Electrical Systems: Technologies for Planning, Operation, and Risk Management (30 hours). Comité Colombiano de la CIER (COCIER), virtual., Jul. 2026

Shipley Introduction to AI for Proposals course, 2025

Powerful Proposal Writing for Postdocs course, 2025

CIRTL Associate Level of Certification. Center for the Integration of Research Teaching and Learning (Online), University of Tennessee, Knoxville, Jun. 2023

Certification in University Teaching. Pontificia Universidad Javeriana, Bogotá, Colombia (Online)., Apr. 2023

Certification in University Teaching and Didactics. Politécnico Superior de Colombia, Medellín, Colombia (Online), Jan. 2023

Energy Leaders Training Program (Programa de Formación de Líderes Energéticos), Edition 13, World Energy Academy (39 hours). World Energy Council Colombia, 2021

SPECIALIZED SKILLS

Electrochemistry: Electrochemical impedance spectroscopy (EIS, GEIS, PEIS) and equivalent-circuit modeling; cyclic voltammetry; linear sweep voltammetry; chronoamperometry; chronopotentiometry; galvanostatic cycling; polarization curves; rotating disc electrode and microelectrodes.

Spectroscopy: UV-Vis; FTIR; NMR (^{1}H, ^{13}C, ^{31}P); Raman; Electron Paramagnetic Resonance (EPR).

Separation and analysis: Gas chromatography; HPLC.

Computation and software: Python and MATLAB; EC-Lab, MestReNova, ChemDraw; Microsoft Office.

PROFESSIONAL SOCIETIES

2024 – present **| The Electrochemical Society (ECS)**, Member.

LANGUAGES

English: professional working proficiency.

Spanish: native.